3AL Test 1 2023 Solutions

Time: 90 minutes **Available marks:** 70 **Full marks:** 65

Instructions

1. Answer all questions in the lined answer booklets.
2. You can answer the questions in any order but start each question on a new page.
3. Make sure each answer is clearly labelled.
4. Show all work and reasoning unless otherwise instructed.
5. This test consists of **4** questions. Make sure your question paper is complete.
6. If you have read the instructions, draw a star on the front of your answer booklet. You will receive an extra mark for doing so.

Good luck! ✧

1. Let G_1 and G_2 be groups. Use the First Isomorphism Theorem to prove that there is a factor group of $G_1 \times G_2$ that is isomorphic to G_1. Clearly show what the factor group is.

   NOTE: This question is asking you to find a normal subgroup $H \triangleleft G_1 \times G_2$ such that $(G_1 \times G_2)/H \cong G_1$.

Consider the set $H = \{(1_{G_1}, g_2) : g_2 \in G_2\} \subseteq G_1 \times G_2$. We claim that $H \triangleleft G_1 \times G_2$ and $(G_1 \times G_2)/H \cong G_1$. We define a map $\varphi : G_1 \times G_2 \to G_1$ by $\varphi(g_1, g_2) = g_1$. For $(g_1, g_2), (g_1', g_2') \in G_1 \times G_2$, we have

$$\varphi((g_1, g_2)(g_1', g_2')) = \varphi(g_1g_1', g_2g_2') = g_1g_1' = \varphi(g_1, g_2)\varphi(g_1', g_2').$$

Thus φ is a homomorphism. Moreover, for any $g_1 \in G_1$, we have $(g_1, 1_{G_2}) \in G_1 \times G_2$ such that $\varphi(g_1, 1_{G_2}) = g_1$. Thus φ is an epimorphism. We have $(g_1, g_2) \in \ker\varphi$ if and only if $\varphi(g_1, g_2) = 1_{G_1}$. That is, $(g_1, g_2) \in \ker\varphi$ if and only if $g_1 = 1_{G_1}$. Thus $\ker\varphi = H$. Then by the First Isomorphism Theorem, $(G_1 \times G_2)/H \cong G_1$ (the normality of H also follows from the theorem).

[15 marks]

2. (a) State the Second Isomorphism Theorem.

(b) Let G be a group of order $p^n q$, where p and q are distinct primes and $n \geq 1$. Let K be a normal subgroup of G of order p^n. Suppose H is a subgroup of G.

(i) Show that $|HK/K|$ divides q.

(ii) Show that if $|H| = p^n$, then $|HK/K| = 1$ and $|H \cap K| = p^n$.

(iii) Explain why K is the only subgroup of G with order p^n.

(a) **Second Isomorphism Theorem.** *Let H and K be subgroups of G with $K \triangleleft G$. Then $H/(H \cap K) \cong (HK)/K$.*

(b) (i) By the Correspondence Theorem, $HK/K \leq G/K$. By Lagrange's Theorem, $|G/K| = p^n q/p^n = q$. Then, by Lagrange's Theorem again, $|HK/K|$ must divide $|G/K| = q$.

(ii) By the Second Isomorphism Theorem, $H/(H \cap K) \cong HK/K$. Thus the orders of these groups must be equal. Since $H \cap K$ is a subgroup of H, its order must divide that of H. That is, $|H \cap K|$ is a power of p. Then $|H|/|H \cap K| = p^k$ for some $0 \leq k \leq n$. Then $|HK/K| = p^k$. But $|HK/K|$ divides q, which is not divisible by p. So, we must have $k = 0$. That is, $|HK/K| = 1$. Moreover, $|H \cap K| = |H|/p^k = |H|/1 = p^n$.

(iii) If there was another subgroup H of G with order p^n, then we would have $|H \cap K| = p^n$. But $H \cap K$ is a subgroup of H, which also has order p^n. Then $H \cap K = H$. Then $H \subseteq K$ but $|H| = |K|$ and so $H = K$. Thus, K is the only subgroup of G with order p^n.

[5 + 5 + 10 + 5 = 25 marks]

3. Let G be a group with normal subgroups H and K such that $H \subseteq K$. Prove or disprove: If G/H is cyclic, then so are G/K and K/H.

Since G/H is cyclic, so are its subgroups. Thus, K/H is cyclic. Moreover, since G/H is cyclic, so are its factor groups. By the Third Isomorphism Theorem, $G/K \cong (G/H)/(K/H)$. Thus $(G/H)/(K/H)$ is cyclic and then so is G/K, since it is isomorphic to the aforementioned group.

[10 marks]

4. Consider the group $G = \mathbb{Z}_{250}$ with subgroup $K = \langle [10] \rangle$. Note that $250 = 2 \cdot 5^3$.

(a) Determine all subgroups of $\mathbb{Z}_{250}$ and draw the lattice diagram.

(b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{250}/K$.

(c) Let $H = \langle [50] \rangle$. Determine the number of subgroups of $(\mathbb{Z}_{250}/H)/(K/H)$.

NOTE: You can use the fact that isomorphic groups have the same number of subgroups (no need to prove this).

(a) By the Fundamental Theorem of Finite Cyclic Groups, the subgroups of $\mathbb{Z}_{250}$ correspond to the divisors of 250. Namely, 1, 2, 5, 10, 25, 50, 125 and 250. The subgroups of $\mathbb{Z}_{250}$ are the unique subgroups groups of these orders. Namely, $\{0\} = \langle 250 \rangle$, $\langle 125 \rangle$, $\langle 50 \rangle$, $\langle 25 \rangle$, $\langle 10 \rangle$, $\langle 5 \rangle$, $\langle 2 \rangle$, $\mathbb{Z}_{250} = \langle 1 \rangle$. The lattice diagram is:

$$\begin{tikzcd}
 & & & \mathbb{Z}_{250} = \langle 1 \rangle \arrow[dl, dash] \arrow[dr, dash] & \\
 & & \langle 5 \rangle \arrow[dl, dash] \arrow[dr, dash] & & \langle 2 \rangle \arrow[dl, dash] \\
 & \langle 25 \rangle \arrow[dl, dash] \arrow[dr, dash] & & K = \langle 10 \rangle \arrow[dl, dash] & \\
\langle 125 \rangle \arrow[dr, dash] & & \langle 50 \rangle \arrow[dl, dash] & & \\
 & \{0\} = \langle 250 \rangle & & &
\end{tikzcd}$$

(b)

$$\begin{tikzcd}
 & \mathbb{Z}_{250}/K \arrow[dl, dash] \arrow[dr, dash] & \\
\langle 5 \rangle / K \arrow[dr, dash] & & \langle 2 \rangle / K \arrow[dl, dash] \\
 & K/K &
\end{tikzcd}$$

(c) By the Third Isomorphism Theorem, $(\mathbb{Z}_{250}/H)/(K/H) \cong \mathbb{Z}_{250}/K$. In the above lattice diagram, we see that $\mathbb{Z}_{250}/K$ has 4 subgroups. Thus, $(\mathbb{Z}_{250}/H)/(K/H)$ also has 4 subgroups.

[10 + 5 + 5 = 20 marks]